Sensor Fusion

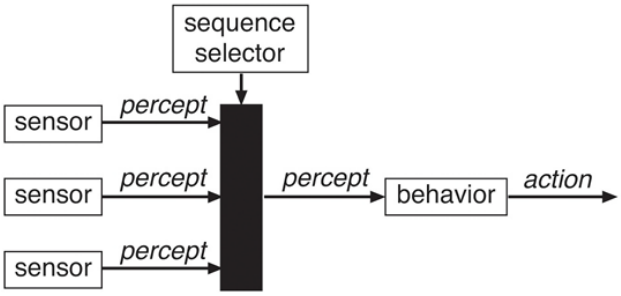
It is a broad term for any process combining information from multiple sensors into a single percept. It means constructing a global world model in a hierarchical or deliberative system.

Basic Combinations of Sensors

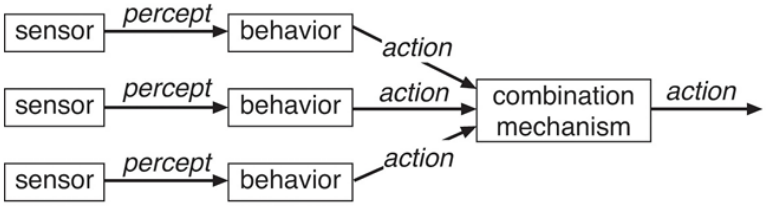
- **Redundant (Competing)** - This type of sensor configuration involves using multiple sensors that measure the same quantity to provide a backup in case one sensor fails.
    - **Physical redundancy** refers to using multiple physical sensors that measure the same quantity.
    - **Logical redundancy** refers to using multiple sensors that measure different quantities but can be used to infer the same information.
- **Complementary** - This type of sensor configuration involves using multiple sensors that measure different aspects of the same phenomenon and then combining the data to improve the accuracy of the measurement.
- **Coordinated** - These sensors use a sequence of sensors, often for cueing or providing focus-of-attention.

Types of Behaviors in Sensors

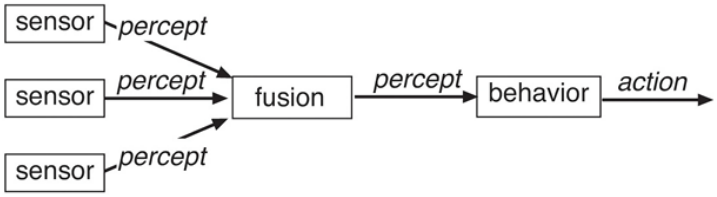
- **Sensor fashion** refers to coordinated sensor usage where a robot employs a sequence of sensors to cue or provide focus-of-attention. **Fashion** implies that the robot changes sensors in response to changing circumstances, similar to how people alter their clothing styles based on the seasons.



- **Sensor fission** is a design philosophy where a robot uses one sensor per behavior or task, with behaviors sharing a sensor stream without awareness. ***Fission*** comes from the connotation of "fusion" in nuclear physics, where energy is generated by separating atoms and particles.



- **Action-oriented sensor fusion** is a type of sensor fusion that converts sensor data into behavior-specific representation to support a specific action rather than constructing a global world model. Multiple sensor streams for behavior can be active and influence the resulting behavior in action-oriented sensor fusion.



Sensing Modalities

These are the methods by which robots can collect information about their environment and their own internal state.

- **Proprioception** is a way for robots to measure their movements and orientation using sensors. For example, a robotic arm can use joint angle sensors to determine the position of each joint.
- **Exteroception** is a way for robots to understand the layout of their surroundings using sensors. Cameras, lidars, and sonars are exteroceptive sensors that can provide information about objects and their distance from the robot.
- **Exproprioception** is a way for robots to determine their position relative to their environment using sensors. For example, a robot's wheel encoder can provide information about how far it has traveled, allowing it to keep track of its position relative to a starting point.

Sensor Suite Attributes

Sensor suite is a group of sensors that work together to measure different physical and environmental variables. In robotics, a sensor suite is often employed to help a robot perceive and interact with its surroundings.

1. **Field of View and Range**
    a. **FOV** refers to the extent of the observable environment that a sensor can capture or monitor at any given time.
    b. Types: ***horizontal FOV*** and ***vertical FOV***
    c. **Range** - the length of the field.
2. **Accuracy, Repeatability, and Resolution**
    a. **Accuracy** refers to the degree to which a sensor's measurement corresponds to the true value of the measured quantity. A highly accurate sensor will produce readings very close to the true value.
    b. **Repeatability** refers to the ability of a sensor to produce the same output for the same input under the same conditions over time. (Consistency)
    c. **Resolution** refers to the smallest change in the measured quantity a sensor can detect and differentiate. (Firmness of Decision)
3. **Responsiveness.** Most sensors have particular environments in which they function poorly. The environment must allow the signal of interest to be extracted from noise and interference.
4. **Power consumption.** It is always a concern how robots, including their sensors, consume the power provided quickly.
    a. **Hotel Load** - The amount of power a mobile robot requires to support a sensor package.
    b. **Locomotion Load** - The power needed to move the robot.
5. **Reliability.** An important parameter is needed for a group of sensors to work in all kinds of environments. Sensors often have physical limitations regarding their work, such as temperature and moisture constraints, which must be considered.
6. **Size.** The size of sensors affects the overall design and integration into robotic systems. Compact and lightweight sensors are often preferred, especially in applications with limited space.
7. **Computational complexity.** It is the estimate of how many operations an algorithm or program performs.

Range Sensing

To function intelligently in the world, robots need to be able to measure or extract depth or range. The terms “depth” and “range” are used synonymously in robotics to refer to the distance between the robot and a surface in the world.

- **Stereopsis** refers to using two or more cameras or sensors to capture slightly different images of the same scene. Example: using two cameras to extract range data. The cameras are known as the *stereo pair*.
- **Depth from X** refers to the process of determining the distance or depth of an object or surface from a particular reference point or axis, denoted as X. The term "X" can refer to a specific camera or sensor, or it can refer to a particular location or coordinate system.
- **Sonar or Ultrasonics** – these sensors use sound to interpret information.
    - **Sonar** is any system that uses sound to measure ranges operating at different frequencies.
    - **Ultrasonics** are active sensors that emit a sound and measure the time it takes for the sound to bounce back.

- **Examples:**
  - **Light Stripers** project a colored line (or stripe), grid, or pattern of dots on the environment.
  - **Lidar** are devices that use lasers to produce a depth map or image.
  - **RGB-D Cameras** combine an IR light striping system with a regular RGB visible light camera. RGB-D cameras project IR light read by a camera tuned for the IR frequency of light.
  - **Point Clouds** are a knowledge representation of the geometry and spatial range of surfaces in a three-dimensional environment. The cloud is a collection of points, usually thousands of points. The cloud can be an array or a list of points.
    - Each point has either a (*x, y, z*) coordinate, called a ***vertex***, representing the distance between the sensor and the surface of an object or a ***vector*** (*direction, distance*).

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